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Summary

Full-stack engineer with expertise in React, MongoDB, and SQL, driven by building solutions to problems I have encountered firsthand. Projects span production MERN systems, serverless AWS architecture, and research-grade XR accessibility tools.

Education

Indian Institute of Information Technology Sricity

August 2023 - June 2027

Bachelor of Technology in Computer Science (CGPA of 7.63)

Projects

HalfBeat 🎧 | *Unity, C#, XR Interaction Toolkit, OpenXR, EzySlice*

2026

Accessible VR rhythm game designed for players with motor disabilities and limb differences.

- Designed from firsthand experience with motor disabilities. Engineered a one-handed VR accessibility system with position mirroring, freeze-offset, and saber-swap modes enabling full gameplay using a single controller.
- Conducted user study with 17 participants; NASA-TLX Raw TLX of 42.0 (low-to-moderate workload) and SUS of 65.8 (marginal usability); 12 of 13 long-session participants showed a negative learning slope, confirming motor adaptation — developed as research artifacts IMAP and NERD targeting UIST 2026.

SplitSmart 🎧 | *MERN, Socket.io, Gemini Vision API, JWT*

2026

AI-powered group expense splitter with real-time cooperative bill claiming and OCR receipt parsing.

- Engineered a 3-phase greedy minimum cash flow algorithm reducing N debts to at most N-1 transactions; integrated Gemini 2.5-Flash via zero-disk in-memory pipeline for receipt OCR; implemented OCC with jitter-based retry backoff to resolve simultaneous item-claim race conditions across Socket.io clients.
- Optimised query performance with compound MongoDB indexes across 4 models, cutting expense doc scans from 6,021 to 15 and raw DB time from 9ms to 1ms; hardened API with role-based JWT refresh flow, sparse unique indexing for frictionless guest onboarding, and rate limiting to prevent token exhaustion.

Bookish 🎧 | *MERN, Redux Toolkit, Stripe, Upstash Redis, Jest*

2026

Full-stack book marketplace connecting buyers, sellers, and admins with role-based dashboards.

- Integrated Upstash Redis cache-aside layer across 11 critical endpoints achieving 84.82% average latency reduction — warm cache hits consistently under 30ms, down from 120-230ms cold DB access.
- Implemented CI/CD pipeline with GitHub Actions running 48 Jest tests across 12 suites on every push; added B-Tree indexing, field projections, and `.lean()` hydration bypass reducing CPU overhead by 35-50%.

PowerPulse 🎧 | *AWS Lambda, DynamoDB, API Gateway, Cognito, Serverless*

2025

Serverless energy analytics system with rule-based and AI-driven insights.

- Architected a fully serverless backend on AWS Lambda, API Gateway, Cognito, and DynamoDB — eliminating idle infrastructure cost and enabling automatic scaling without capacity planning.
- Devised caching strategy that cut AI response time from 1800ms to 200ms (89% faster) and reduced API costs by 70%; integrated Hugging Face inference with fallback control.

Technical Skills

Languages: C, C++, C#, Java, JavaScript, HTML, CSS

Frontend: React, Tailwind CSS, Redux Toolkit, TanStack Query

Backend: Node.js, Express.js, REST APIs, Socket.io, JWT, Multer

Cloud & Infra: AWS Lambda, API Gateway, Cognito, Cloudinary, Render, Vercel

Databases: MongoDB, MySQL, DynamoDB, Upstash Redis

Tools: Git, GitHub, Jest, GitHub Actions, Unity, OpenXR, XR Interaction Toolkit, Postman

Achievements

- Secured 3rd place at Global Game Jam 2025 — shipped a complete game under a 48-hour constraint.
- Rated 1580 on LeetCode; solved 250+ problems across LeetCode and Codeforces.
- XR Lead, IOTA Club — leading XR initiatives, mentoring members, and managing lab projects.
- HCI Lab — developed IMAP and NERD research artifacts in VR accessibility; targeting UIST 2026.